

AVINASH MISHRA

**AI-First Senior Technical Program Manager | AI Platform Engineering & Closed-Loop AI Evaluation Leadership
| GenAI/LLM Platform Delivery | AI Governance & Cloud FinOps**

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Available for U.S.-based remote roles - Full-Time or Contract | No visa sponsorship required | Full workday overlap with U.S. business hours

25,000+ Active Users | >90% RAG Faithfulness | 99.95% Uptime | 2-3x AI-Assisted Engineering Output | 20-30% Cloud Cost Savings

PROFESSIONAL SUMMARY

AI-first Senior Technical Program Manager with 10+ years directing enterprise SaaS and, most recently, AI platform engineering and GenAI/LLM platform delivery for a multi-tenant system serving 22 institutional clients and 25,000+ end users. Leads by putting AI-assisted engineering, AI workflow automation, and AI governance at the center of program execution - using modern AI capabilities to compound engineering velocity rather than relying on hands-on coding depth. Owns program-level closed-loop AI evaluation: governing the iterative evaluation and continuous-optimization workflow (Ragas, G-Eval, LLM-as-a-Judge; golden datasets; quality gates) that sustains >90% output faithfulness, and driving enterprise-wide AI adoption and governance standards across the engineering organization. Also directed a semantic-caching architecture that cut inference latency and cost, and a bi-weekly FinOps governance model - partnering closely with engineering leadership on AI systems architecture and technical trade-offs while owning roadmap, risk, delivery telemetry (DORA), and executive/customer communication across a 40-45 person distributed engineering matrix. Seeking to bring this AI-enabled delivery discipline to a larger global engineering organization.

CORE COMPETENCIES

AI Leadership: AI Platform Engineering, Agent Orchestration (Agentic AI, Multi-Agent Systems), Closed-Loop AI Evaluation, AI Evaluation & Observability, Context Engineering, Model Context Protocol (MCP), AI-Assisted SDLC, AI Workflow Automation, AI Engineering Productivity, AI Governance & Responsible AI, LLMops, Prompt Engineering, Multi-Model Routing, AI Cost Optimization / AI Economics, AI Center of Excellence (CoE), Enterprise AI Adoption, AI ROI & Business Value Realization, Azure AI Foundry

Program Leadership: Technical Program Management, Program & Portfolio Strategy, Executive Stakeholder Management, Cross-Functional Leadership, Global/Distributed Team Leadership, Roadmap Planning, Risk & Dependency Management, Change Management, Release Governance, Agile Delivery & Scrum, DORA Metrics

Cloud & Platform: AI Systems Architecture, Distributed Systems, Scalability & Reliability, Enterprise SaaS, Multi-Tenant Architecture, Cloud FinOps, AWS, Azure, GCP, Kubernetes, Microservices, REST APIs

Engineering Productivity: Cursor, GitHub Copilot, Sourcegraph Cody, Amplitude, AI-Assisted Engineering, Delivery Analytics

PROFESSIONAL EXPERIENCE

Senior Technical Program Manager & Founder - Noetic Education Pvt. Ltd.

Jun 2016 - Present | Mumbai, India

Directs a bootstrapped, self-funded B2B enterprise EdTech platform (founded 2016), governing 5-8 concurrent programs and a 40-45 person distributed engineering matrix delivering 60+ software systems for 22 enterprise clients, 500+ business users and 25,000+ active end users. Platforms are deployed white-label under each client's own brand by design - protecting institutional goodwill rather than building a public consumer brand - including a premium white-label add-on, StudyAtom, licensed within the existing client base.

Enterprise GenAI & RAG Transformation | 2024 - 2026

- AI Platform Engineering & Agent Orchestration:** directed the program establishing GenAI/LLM capability as a reusable, model-agnostic AI platform and agent orchestration layer - rather than isolated features - across the full multi-tenant client base, with capability-based multi-model routing (Claude Haiku/Sonnet, GPT-4o) and automatic failover on provider rate-limits.
- Closed-Loop AI Evaluation & Context Engineering:** owned program governance of a closed-loop, iterative AI evaluation and continuous-optimization workflow - defining AI quality gates, governing the golden dataset and LLM-as-a-Judge evaluation program (Ragas, G-Eval) gated in CI/CD, and directing context engineering optimizations across prompt, retrieval, and model-routing layers from evaluation results - measuring faithfulness, context precision and answer relevance to enforce production-readiness before release; sustained >90% output faithfulness, with monthly Cohen's Kappa calibration (>0.8) to control judge bias.
- Latency & cost engineering:** set technical strategy for a semantic-caching architecture (vector similarity) achieving a 30-40% cache-hit rate, cached TTFT p95 <500ms and ITL p95 <50ms, including directing root-cause analysis with engineering on an SSE-buffering load-balancer defect.

- **Responsible AI:** drove adoption of PII-tokenization and mandatory source-citation guardrails into the SDLC, turning production AI output into auditable, source-grounded answers rather than unverified text.
- **AI Cost Optimization / AI Economics:** governed model routing, semantic caching and token-usage optimization as commercial-sustainability decisions via bi-weekly FinOps reviews - identifying 20-30% cloud cost savings and >60% spot/savings-plan utilization, with per-feature/tenant unit-economics reporting down to \$0.15 per processed transaction for a high-volume feature.
- **Delivery telemetry:** instituted DORA-based metrics (commit lead time <24-48 hrs, change failure rate <15%, 2-4 week feature lead time) and a 70/20/10 features/tech-debt/maintenance capacity model.
- **AI-assisted engineering productivity:** drove team-wide rollout of AI-assisted coding tools (Cursor, GitHub Copilot, Sourcegraph Cody) across the engineering matrix by 2025 - an AI-first workflow-automation initiative that lifted overall engineering output 2-3x and compounded the DORA velocity gains above.

Premium Add-On Productization - StudyAtom (White-Label) | 2022 - 2023

- **Premium upsell line:** directed development and rollout of StudyAtom, a white-label digital-learning add-on (course catalogue, content delivery, assessments, progress tracking) licensed as a premium tier to the existing 22-client base - expanding wallet share without a public consumer brand or channel conflict with clients.
- **Architecture governance:** partnered with engineering leadership to shape the decision to decouple the add-on module from the core ERP/LMS into a shared headless microservices backend with independent deployment pipelines - improving add-on engineering velocity (up to 5 deploys/day) while preserving enterprise stability and QA-gated bi-weekly release governance for core systems.
- **Governance maturity:** matured program governance (quarterly roadmap planning, cross-functional dependency tracking, executive reviews) spanning Engineering, Product, Academic Content, Client Success and Support.

High-Availability Platform Scaling | 2019 - 2021

- **Scalability & Reliability:** directed platform-wide scaling through COVID-driven demand surge, growing the platform from 5 to 60+ software systems and from ~2,000 to 25,000+ active end users across 22 institutional clients, while sustaining a 99.95% availability target.
- **Distributed Systems Reliability:** directed cross-functional incident response and root-cause analysis on database connection-limit exhaustion under concurrent-user spikes, prioritizing engineering effort and governing rollout planning for a zero-downtime remediation using centralized connection pooling (PgBouncer) and asynchronous read-replica offload.
- **Unified operations:** directed consolidation of disconnected academic workflows (admissions, attendance, exams, communication) into one digital platform, governing release planning across multiple concurrent releases delivered with no service degradation.

Enterprise Platform Modernization & Multi-Tenant SaaS Migration | 2016 - 2018

- **Program leadership:** directed a multi-year platform modernization program transitioning fragmented, spreadsheet-driven institutional workflows (admissions, attendance, fees, exams) into a unified, multi-tenant SaaS platform - replacing per-client custom builds with one reusable foundation; grew the initial engineering function to deliver the first 5 core systems, supporting up to 2,000 active end users.
- **Architecture direction:** set architecture direction for institution-level data isolation, configurable workflows and API-first integrations, establishing the reusable platform foundation that scaled 12x in systems and users during the COVID growth phase.
- **Agile governance:** introduced structured Agile delivery (sprint planning, backlog refinement, risk/dependency reviews) across the engineering function, sustaining an 80-85% sprint say-do ratio.

TECHNICAL FLUENCY

AI / GenAI: Agentic AI, Agent Orchestration, AI Evaluation, Closed-Loop AI Evaluation, Context Engineering, Model Context Protocol (MCP), Azure AI Foundry, LLM Orchestration & Multi-Model Routing, RAG Architecture, Prompt Engineering, Ragas, G-Eval, LLM-as-a-Judge, Semantic Caching, Vector Search, Anthropic API, OpenAI API (model routing & evaluation)

Cloud & Infrastructure: AWS, Azure, GCP, Firebase, PostgreSQL, MySQL, PgBouncer, Read Replicas, Spot/Savings Plans, Kubernetes, Microservices & REST APIs

Delivery & Governance: Agile, Scrum, Kanban, DORA Metrics, Multi-Tenant SaaS Architecture, Release & Risk Management, Cloud FinOps (CAPEX/OPEX)

AI-Assisted Engineering: Cursor, GitHub Copilot, Sourcegraph Cody, Amplitude, AI Workflow Automation, Delivery Analytics

Collaboration Tools: Jira, Confluence, Git, GitHub, GitLab, Slack, Microsoft Teams, Notion

EDUCATION

Bachelor of Science, Industrial Science - 2013

Aircraft Maintenance Engineering (Avionics) - 2013